



SEQUOIA IMPROVED COOKSTOVES PROJECT IN TOGO



Document Prepared by Aera Group

On behalf of

SEQUOIA PLANTATION TOGO SA U

Project Title	Sequoia Improved cookstoves project in Togo
Version	1.8
Date of Issue	30/05/2024
Prepared By	Aera Group
Contact	Address: 28 cours Albert 1er, 75008 Paris, France

CONTENTS

1	PROJECT DETAILS.....	4
1.1	Summary Description of the Project	4
1.2	Sectoral Scope and Project Type	5
1.3	Project Eligibility	5
1.4	Project Design	7
1.5	Project Proponent	8
1.6	Other Entities Involved in the Project	8
1.7	Ownership.....	9
1.8	Project Start Date	9
1.9	Project Crediting Period	9
1.10	Project Scale and Estimated GHG Emission Reductions or Removals	9
1.11	Description of the Project Activity	10
1.12	Project Location	13
1.13	Conditions Prior to Project Initiation	14
1.14	Compliance with Laws, Statutes and Other Regulatory Frameworks	14
1.15	Participation under Other GHG Programs	14
1.16	Other Forms of Credit.....	15
1.17	Sustainable Development Contributions	15
1.18	Additional Information Relevant to the Project	16
2	SAFEGUARDS	26
2.1	No Net Harm	17
2.2	Local Stakeholder Consultation	17
2.3	Environmental Impact	25
2.4	Public Comments	25
2.5	AFOLU-Specific Safeguards	25
3	APPLICATION OF METHODOLOGY.....	35
3.1	Title and Reference of Methodology	25
3.2	Applicability of Methodology	25
3.3	Project Boundary	27
3.4	Baseline Scenario	28

3.5	Additionality	28
3.6	Methodology Deviations	29
4	QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	40
4.1	Baseline Emissions	29
4.2	Project Emissions	30
4.3	Leakage.....	30
4.4	Net GHG Emission Reductions and Removals	30
5	MONITORING	48
5.1	Data and Parameters Available at Validation	39
5.2	Data and Parameters Monitored.....	43
5.3	Monitoring Plan.....	48

1 PROJECT DETAILS

1.1 Summary Description of the Project

The project activity aims to provide individuals with intermediate improved cookstove (ICS). The project is taking place in Togo . The stoves will be distributed for free in all the regions of Togo: Maritime, Plateaux, Centrale, Kara, Savanes..

The ICS distributed through this project will encourage households to replace the prevailing inefficient three-stone fireplace or traditional charcoal cookstoves such as charcoal clay cookers and charcoal iron stoves that are predominantly used across the region. The ICS provides for an efficient use of thermal energy generated through the combustion of biomass while cooking and there is a more efficient transfer of this thermal energy to the cooking pot. There is thus a resultant reduction in both biomass consumption and greenhouse gas emissions.

In addition to the reduction of biomass consumption (and associated reduction in expenses to purchase wood or charcoal – or reduction in time gathering firewood), there will also be an economic benefit in that the ICS will be produced locally in the region and will create both direct and indirect jobs related to the production, the distribution and the maintenance of the cookstoves.

Since 2015, the Togolese population continued to increase (more than a million more inhabitants) meaning a constant increase of wood fuel consumption. As per latest available data from Forest resource assessment¹ forest area continues to decrease (-59,190 ha between 2000 and 2020) with a deforestation rate of 4,960 ha/year. At the same time, biofuel is still the main fuel source used by households for energy².

The project is expected to avoid the emissions of 5,796,120 tCO₂e during the first crediting period and on average 828,017 tCO₂e per year.

Audit Type	Period	Program	VVB Name	Number of years
Validation (crediting period)	18-09-2022 – 17-09-2029	VCS	EARTHOOD SERVICES PRIVATE LIMITED	7 years (renewable)

¹ <https://fra-data.fao.org/assessments/fra/2020/TGO/sections/extentOfForest>

² <https://au-afrec.org/togo>

Total	-	1	1	-
-------	---	---	---	---

1.2 Sectoral Scope and Project Type

Sectoral Scope: 03

Project Type: Type II –Energy Efficiency Improvement Project

Grouped Project: No.

1.3 Project Eligibility

As per para. 2.1.1 of the VCS Standard, V.4.4 the project is eligible under the scope of the VCS Program because the VCS Scope includes project activities involving the following elements:

Elements	Project applicability
The six Kyoto Protocol greenhouse gases	The project activity aims at CO2 reduction of non-renewable biomass.
Ozone-depleting substances.	n/a
Project activities supported by a methodology approved under the VCS Program through the methodology approval process.	The project applies VMR0006 v1.1 which is a VCS methodology.
Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval.	n/a
Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements.	n/a

The project is not also part of any excluded VCS activities.

Excluded Activity	Applicability
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	N/A. The GP reduces GHG emissions associated with the combustion of cooking fuels

Grid-connected electricity generation using hydro-power plants/units	N/A The project activity are energy efficiency measures and do not generate electricity from hydro-power plants/units.
Grid-connected electricity generation using wind, geothermal, or solar power plants/units	N/A. The project activity is energy efficiency measures and do not generate electricity from wind, geothermal, or solar power plants/units.
Utilization of recovered waste heat for, inter alia combined cycle electricity generation and the provision of heat for residential, commercial, or industrial use	N/A. The project activity is energy efficiency measure and do not utilize recovered waste heat for any purposes.
Generation of electricity and/or thermal energy using biomass. This does not include efficiency improvements in thermal applications (e.g., cookstoves)	N/A. The project activity is energy efficiency measures and do not generate electricity using biomass. The GP includes energy efficiency improvements in thermal applications, i.e. cookstoves.
Generation of electricity and/or thermal energy using fossil fuels, including activities that involve switching from a higher carbon content fuel to a lower carbon content fuel	N/A. The project activity does not include the generation of electricity and/or thermal energy using fossil fuels, including activities that involve switching from a higher carbon content fuel to a lower carbon content fuel.
Replacement of electric lighting with more energy efficient electric lighting such as the replacement of incandescent electrical bulbs with CFLs or LEDs	N/A. The project activity does not include the replacement of electric lighting.

Installation and/or replacement of electricity transmission lines and/or energy efficient transformers	N/A. The project activity does not include the installation and/or replacement of electricity transmission lines and/or energy efficient transformers.
--	--

Furthermore, it cannot reasonably be assumed that the project generates GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction. All the households involved in this project activity are fully independent from the project proponent.

The project aims to provide individuals with efficient or improved cookstove (ICS) which falls under the category of efficiency improvements in thermal applications, therefore it is eligible under the scope of VCS Program. As per VCS Standard Version 4.4, efficiency improvements in thermal applications (e.g., cook stoves) are not excluded, hence project is eligible under VCS.

The stoves will be distributed into the regions via free distribution. Double counting will be prevented by each ICS distributed as part of this project activity will have a project identifier and unique serial number inscribed in the metal. This will prevent any other project proponent, organization, retailer, individual or distributors from using the same ICS and including it in another project activity

1.4 Project Design

The project activity involves the free distribution of ICS devices to households within the project boundary (Togo) ICS are distributed for free to households in all the region of Togo (Maritime, Plateau, Centrale, Kara Savanes). ICS will be distributed to the population that are still using inefficient cookstoves. All the stoves distributed have a unique serial number, and all the cookstoves distributed will be recorded in an electronic

Excel database, which will be checked to ensure that double counting is avoided. Furthermore, the project will target end-users in the project boundary that are still using inefficient cookstoves. The project database will be available to VVB to confirm that all the cookstoves are distributed in the previously stated regions (Maritime, Plateau, Centrale, Kara, Savanes). Cookstoves distributed outside of the project boundary, will not be accounted for.

Moreover, the projects aim at distributing only one ICS per household. Even if more than 1 ICS is found in a household, only one will be accounted for ER calculations.

To account for the continued use of baseline devices, the adjustment factor will be assessed during monitoring period to calculate a Boldadjusted as per methodology requirements.

The distribution of energy efficient cookstove distribution falls under the category of efficiency improvements in thermal applications of non-renewable biomass, therefore it is eligible under the scope of VCS Program

Eligibility Criteria

The project is a not a grouped project hence this is not applicable.

1.5 Project Proponent

Organization name	Sequoia Plantation Togo SAU
Contact person	Aavaneesh Arja
Title	General Manager
Address	AGOE, rue CACAVELI, BP 12917, Lome 02, Togo
Telephone	+22870590352
Email	aavaneesh.arja@arisenet.com

1.6 Other Entities Involved in the Project

Organization name	Aera Group
Role in the project	Project advisors
Contact person	Alexandre Dunod
Title	COO
Address	28 cours Albert 1er - 75008 Paris
Telephone	+33 6 42 96 09 78

Email	a.dunod@aera-group.fr
-------	--

1.7 Ownership

Sequoia Plantation Togo SAU the project owner, has granted full and uncontested legal ownership of the emission reductions that are generated under this VERRA project to Aera Group SAS, and has legal rights concerning changes in use of resources required to service the Project for ownership of emission reductions.

Each ICS has a unique serial number linking it to the project activity and is necessary to establish the fact that the ICS belongs to this VCS project. End users will be informed at the time of ICS installation/distribution that the ICS are distributed for free on account of them generating carbon credits and the fact that these credits belong to project proponent. The end user will be required to sign an agreement that transfers the rights of emissions reduction generated by their cookstoves to Sequoia Plantation Togo SAU.

1.8 Project Start Date

Project start date is 18-September-2022. As per project database, it is the date of the first ICS distributed & start of GHG emissions reductions³.

1.9 Project Crediting Period

The project crediting period will be seven (7) years – renewable 2 times for a total of 21 years maximum. Starting from 18-September-2022, ending 17-September-2029 for Togo.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	
Large project	x

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
------	--

³ As per VCS standard V4.4; “The project start date of a non-AFOLU project is the date on which the project began generating GHG

emission reductions or carbon dioxide removals”.

18/09/2022-31/12/2022	2,089
2023	360,865
2024	715,365
2025	797,553
2026	878,052
2027	1,043,823
2028	1,171,735
01/01/2029-17/09/2029	826,638
Total estimated ERs	5,796,120
Total number of crediting years	7
Average annual ERs	828,017

1.11 Description of the Project Activity

Several types of ICS models will be distributed in the project. All the models distributed will verify the applicability conditions and same will be verified by VVB. For now, two ICS models are distributed in Togo. The models distributed so far are the model “TCHA TCHA Business” and the model “GLOIRE A DIEU”. They are produced from moulded clay which is dried through high temperatures in a kiln to reinforce its properties including high thermos-efficiency and durability, or alternatively dried in a cool, dry, ventilated area out of direct sunlight to cure the final products. It is a simple to use, multi-purpose stove that can be used in a clay, metal cladded or even built into a hearth of a home for more permanent use. Those models are distributed to households, in both rural and urban areas and use exclusively charcoal. Both models use exclusively charcoal as fuel and have similar characteristics as presented in the table below

Table 1: Specifications of the stoves

GLOIRE A DIEU MODEL	TCHA TCHA Business MODEL
---------------------	--------------------------

Weight: about 12 kg	Weight: about 12 kg
Dimensions: 24 cm high x 30 cm wide	Dimensions: 24 cm high x 30 cm wide
Cooking power: 3.615 kW	Cooking power: 3.615 kW
Efficiency: > 31%	Efficiency: > 31%
Durability: 5 years	Durability: 5 years
Fuel: charcoal	Fuel: charcoal
Type of household: mobile	Type of household: mobile

Figure 1: Pictures of the stoves



Data collection of ICS end-user

Project proponent must gather the necessary information to identify households using its ICS during the course of the project. To facilitate this process, each ICS will be assigned a unique serial number. This number will be recorded during the registration process together with the following information (as appropriate and as available):

- Name of ICS user or head of the household
- Address / GPS of ICS household
- Phone number of ICS user or household, where available
- Stove model
- Date of distribution/installation
- ICS serial number
- Type of fuel used.

The information collected will be stored for 2 years in the electronic database which will serve as project database for project monitoring and sampling purposes.

The database will be checked periodically and only one stove per household will be accounted for emission reduction calculations

The ICS distributed so far “TCHA TCHA Business and GLOIRE A DIEU” have a better efficiency than traditional firewood or charcoal stoves.

TCHA TCHA Business and GLOIRE A DIEU have efficiency of 31.82% thus avoiding GHG emissions linked to the combustion of fuels. The project aims to avoid on average 828,017 tCO₂e per year and 5,796,120 tCO₂e during the first crediting period.

The project will target households in both rural and urban using whether Firewood or charcoal-based stoves.

Only ICS distributed as replacement of inefficient charcoal or firewood stoves will be accounted for emissions reduction calculation. To prevent double counting, each ICS is uniquely identified by the stoves plate (see pictures of figure 1) , that encompass:

- 1) The name of the project developer: Sequoia Plantation
- 2) The country where the project is taking place : Togo serial number:
- 3) The serial number. Each stove is identified as TGXXXXX with:
 - a. TG for Togo

- b. XXXXXX: being the serial number of the stove.

1.12 Project Location

The project activity will be located within the geographic borders of the republic of Togo. The distribution will occur within all the regions of Togo namely regions of Maritime, Plateaux, Centrale, Kara, Savanes.

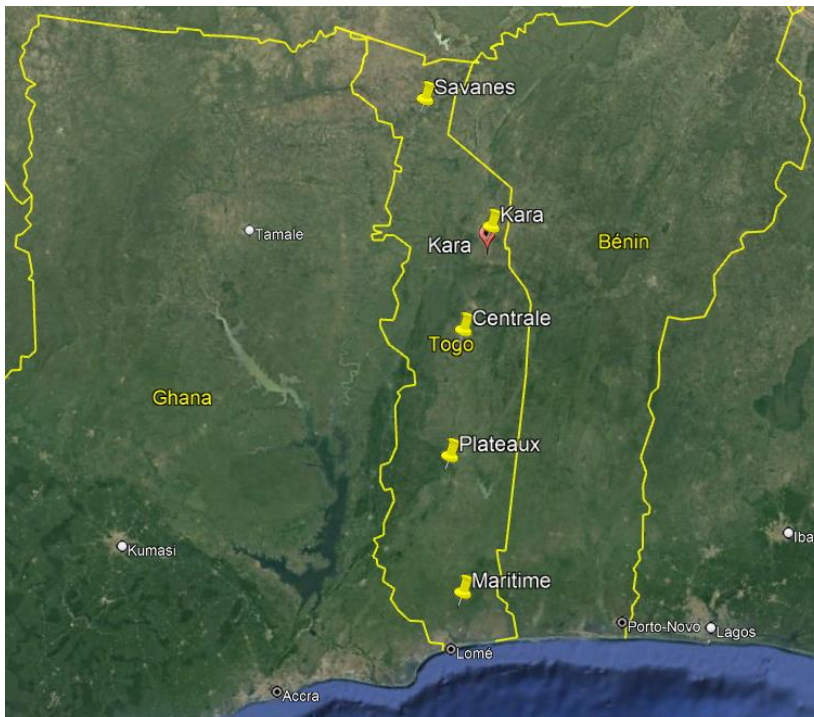


Figure 3: Map of Togo's regions

Togo regions GPS coordinates:

Maritime : 6° 30'0.00" N 1° 19'59.99" E

Plateau : 7° 30'0.00" N 1° 10'0.01" E

Centrale : 8° 40'0.01" N 1° 00'0.00" E

Kara : 9° 40'0.01" N 0° 55'0.01" E

Savanes: 10° 30'0" N 0° 30'0" E

The project will target rural households in those regions.. Project proponent will ensure that cookstoves are distributed to households in the project boundaries..

1.13 Conditions Prior to Project Initiation

The majority of households in Togo still rely on woody biomass for cooking. Most households use the charcoal cookstoves or three-stone-fire as their principal method of cooking. The condition prior to the project initiation is the same as the baseline scenario in section 3.4

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project aligns with the labour convention and labour rights of Togo which is a member of the International Labour Organisation (ILO) since 07.06.1960 and is adhering to all fundamental ILO convention. The project will also align with all the condition in any other country where it will take place. Apart from that, the project activity does not have any regulatory requirements in Togo, as a review of relevant laws of Togo did not highlight any specific concerns or requirements around cookstoves or any aspects related to this project activity. Same will be check for each new country included. Therefore, all activities associated with the project will be undertaken in compliance with local business and social regulations and requirements..

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project is not seeking registration in whole or in part under any other GHG program outside this project activity within VERRA.

1.15.2 Projects Rejected by Other GHG Programs

The project is not seeking registration in whole or in part under any other GHG program outside this project activity within VERRA, and no components have been rejected or excluded from any other GHG emission reduction project.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project activity has no components included in any emissions trading program.

1.16.2 Other Forms of Environmental Credit

The project is not seeking registration in whole or in part under any other form of environmental credit.

List of all other programs under which the project is eligible to participate: GS4GG, CDM, GCC

Supply Chain (Scope 3) Emissions

As per “*DECEMBER 2022 VCS PROGRAM UPDATE*”, the required statement for all emission reductions and removals associated with impacted goods or services will be effective 1 July 2023. Thus, it is not applicable.

1.17 Sustainable Development Contributions

The project activity has a positive impact on sustainable development as outlines below:

ENVIRONMENTAL

Greenhouse gas emissions are reduced by the project activity. This is also aligned to the Sustainable Development Goals (SDG).

Fuel consumption is reduced for cooking, this reduces the amount of non-renewable biomass use in the region. The project will avoid 5,796,120 tCO₂e during the first crediting period (SDG 13.0).

Participating households will be more energy sufficient due to the reduction in fuel biomass requirement due to the efficiency of the devices.

Distribute 680,000 high-efficiency, clean burning Cooktoves in Togo, to benefit over 680,000 households. (SDG 7.1.2)

Significantly reduce Household Air Pollution (HAP) in rural households (SDG 3.9.1).

Thanks to the efficiency of the ICS, wood will be saved and forest areas will be preserved from deforestation (SDG 15)

ECONOMIC

In a region where economic activity and inputs are strained, the annual spend on fuel for cooking will be reduced due to the efficient cookstove use, benefiting households to allow the expenditure of resources elsewhere.

Production of the cookstoves regionally will create employment in the region and promote indirect opportunities in the sales, distribution, training and management roles within the supply chain.

Additional time will be freed in households from fuel collection – these can now be applied elsewhere in economic or social pursuits.

SOCIAL

Reduce the proportion of time rural women spend on unpaid domestic work related to collection of firewood, and cooking meals on inefficient traditional cookstoves. (SDG 5.4.1). Therefore, additional time will be freed in households from fuel collection – these can now be applied elsewhere in economic or social pursuits.

Indoor pollutants – specifically VOCs and particulates are reduced due to more efficient combustion having a positive effect on health. This has been confirmed by the World Health Organisation⁴.

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage.

Commercially Sensitive Information

No commercially sensitive information directly related to the project activity has been excluded from the public version of the project description.

Further Information

No further information was deemed to be required for this project activity.

⁴<https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health>

2 SAFEGUARDS

2.1 No Net Harm

No potential negative environmental or socio-economic impacts have been identified for the project.

2.2 Local Stakeholder Consultation

Public local stakeholder consultation has taken place on October 7, 2022. The public consultation was held in the meeting room of the auditorium of Kkpelé -Adeta. consultation was held following the respect of the different stages of preparations in accordance with the requirements of IFC, the environmental and social framework (CES) of the World Bank.

The consultation was organised by LTA Conseils⁵. LTA CONSEILS is an environmental and social consultancy organisation well implemented in Togo. With regard to their knowledge of Togo regions and population, as well as their expertise in environmental and social studies, they have been contracted by Sequoia to help organize the consultation.

During the consultation the project as well as the environmental and socio-economic benefits of using improved cookstoves were presented.

In total 110 stakeholders from various institutions and beneficiaries attended to the meeting. These actors are composed of local authorities (presenting the Prefect of Kpélélé, the representative of the Mayor of Kpélélé 1); representatives of state institutions such as the ministries of environment, social affairs, local chiefs, opinion leaders, beneficiaries, SEQUOIA PLANTATION managers, NGO managers and the LTA team.

The stakeholders were informed about the local stakeholder consultation by letters written in French and phone calls to remind about the consultation. LTA conseils have identified all the different groups of stakeholders that may be directly or indirectly affected by the project. This includes local and national authorities, NGOs, associations of womens, villages and cantonal chief. The full and detailed list of persons invited is in the stakeholder report⁶.

The invitations were made 1 month before the meeting, on September 7, 2022.

The ceremony was marked by the opening remarks of the representative of the regional director of the environment and forestry resources of the Plateaux region, Lieutenant AGBOKLATCHI Kossivi, in charge of studies and planning at the Regional Directorate for the Environment, who

⁵ <https://ltaconseils.com/>

⁶ Stakeholder consultation report available at : <https://aera.cloudvault.m-files.com/SharedLinks.aspx?accesskey=a6dead71378c5e76968073e42b651c00424e755215a8505b53c4f424be07c9e8&VaultGUID=ffebd84a-a981-4720-85e2-a323c816e47e>

welcomed all the participants and presented the context of this consultation. He also congratulated this initiative which is in line with the Togolese government's directives in terms of its reforestation plan of one billion trees, its commitments under the Paris Agreement and the various actions carried out in favour of the management of the national forest heritage.

His intervention was followed by the opening remarks of the president of the organisation and management committee of the Kpélé Tutu site. He thanked the Togolese government and the financial and technical partners who spare no effort so that Togo can achieve the objectives assigned in the NDP and the government's 2022-2025 roadmap.

The presentation of the project was then made by the director of the consultancy firm LTA Conseils. This presentation covered points such as

- The main proponents of the project which are SEQUOIA PLANTATION and AERA;
- Objective of the consultation - Presentation of the different aspects of the project, notably the resources generated;
- Choice of stove variants based on a national survey on the use of stoves in households
- Characteristics of the selected improved stove including its cost and performance
- Area of implementation of the project which is the whole Togolese territory
- Description of the current situation
- Description of the baseline and project scenarios
- How can carbon finance help? - How will carbon finance help the project?
- Project benefit/ Emissions reduction
- Sustainable development objectives taken into account and safeguard principles.

After the presentation of the project, the organisers provided answers to the stratification of all participants.

Points raised by different groups during the Local Stakeholder and answers are provided below:

Stakeholder	Point raised	Response
Administrative and technical services Direction (Regional Environment and Resources Forestry)	Climate change in Togo, Degradation of the Vegetation. Impact reduction factors Project expectations.	The savannah, Kara, plateaux and maritime regions are subject to the phenomena of climate change through the irregularity of the rains, the shift of the seasons, the wildfires, the anarchic

		carbonization with archaic stoves, etc. The project for the distribution of improved stoves is a commendable initiative that will contribute to the reduction of the use of wood energy and thus to the destruction of plant resources
All actors (Group of women, men, chiefdoms, CCD, CVD, CDQ, Town Hall, Prefecture, Administrations, NGOs and CSOs)	Were you already aware of the project? If so, can you tell us a few words about the project?	Yes, the stakeholders say they learned about the project through the SEQUOIA project coordinator and the firm LTA-Conseils through the invitation letters in which the project was briefly explained as well as its objectives. For stakeholders, the project consists of the free distribution of improved cookstoves. The project is to help reduce pressure on forest species. The project has environmental and socio-economic benefits related to the use of improved stoves.
	In your opinion, what are the objectives of the project and the expected results?	The objective of the project is to help reduce pressure on forest species. The project has environmental and socio-economic benefits related to the use of improved stoves. The results that the project expects is the continued use of the affected stoves as well as its regular maintenance, the reduction of charcoal consumption.

	How do you think you will be involved in the project?	We believe to be involved in the project through the recruitment of local labour in the manufacturing and distribution phases of improved cookstoves.
	Have you ever been involved in previous projects? If so, how was it done?	Yes, we were involved in REDD+ projects to train sheet metal workers on the manufacture of improved cookstoves.
	In your opinion, who are the main actors to be involved in the realization of the project?	The prefectural and regional directorates of the Ministry of Environment and Forest Resources, the directorates of social affairs, the Town Halls, the CCD, CVD, CDQ, chiefdoms as well as the beneficiaries.
	In your opinion, what are the positives/advantages of this project?	The project helps reduce pressure on vegetation cover, reduce GHG emissions and promote good rainfall. On the socio-economic level, the project saves money on the use of improved stoves, avoids skin burns
	In your opinion, what are the negative points of this project?	Stakeholders say that the project does not have any negative impact on both the environment and society.
	What are your expectations of the promoter and the project?	They want all households to benefit from improved cookstoves.
	Would you like to be informed about the progress of the project? And if so, which	They wish to be informed about the progress of the project via public meetings.

	communication channel do you think is the most suitable? (Radio, television, mail, billboard, public meeting, crier, etc.)	
	What are your recommendations for a good success of this project for the benefit of all?	- Educate, inform, and communicate more about the importance of using improved stoves, - Collaborate with stakeholders in the implementation of the project, - Recruit local labour in the implementation of the project, - Provide an improved wood-burning fireplace for households that do not use charcoal.
Women and girls in and out of school	Business Disruption Disturbances of habits and customs	If the change of culinary tools in households will lead to the disruption of cultural and religious principles, it will be necessary to Sensitize men and women on the use of improved stoves.
	Knowledge, management, and risk factors related to gender-based violence (GBV), Violence against children (VCE); Sexual harassment (HS) Exploitation and Abuse Sexual (EAS) Exclusion of vulnerable persons	If the presence of foreign staff, those involved in the distribution of stoves on the one hand, and on the other hand, young men benefiting to the detriment of women and girls in the environment will increase the risks of GBV, EAS and HS, it is necessary to sensitize stove distributors and beneficiary households

		on GBV, VCE and EAS/HS. And have all SEQUOIA workers involved sign codes of conduct. The involvement of women could lead to difficulties in their households and cause violence. Thus, women and girls will have to be more involved in the distribution of stoves. The recruitment of local labour can lead to discrimination that can lead to GBV cases. Thus, a complaint reporting, and management mechanism should be established.
Representatives of civil society (BEE Youth organizations, Women's organization)	Gender-based violence (GBV), Violence against children (VCE); Sexual harassment (HS) Exploitation and Abuse Sexual (SEA) and Exclusion of Vulnerable Persons	Awareness-raising campaigns and distribution of improved cookstoves are likely to cause Gender-Based Violence (GBV), Violence against Children (VAC, Sexual Harassment (HS), Sexual Exploitation and Abuse (SEA) within households. That's why a code of conduct and an intervention action plan taking into account all social strata should be developed. Also, involve NGOs in the prevention and response against GBV, VCE, EAS/HS; Women and girls living with physical disabilities are at risk of being marginalized

		thus vulnerable and elderly people should be favoured when distributing stoves
--	--	--

The outcome of the LSC meeting was more than encouraging, the different stakeholders encouraged and supported the project and wished that the distribution would mainly take into account rural and semi-urban households that do not have the capacity to afford cooking gas to replace the use of wood and charcoal.

None of the queries or comments raised by stakeholders (above), required any design change. No negative impact was detected. All the questions that arose during the meeting were answered back satisfactorily.

As part of the on-going communication and feedback collection on the use of the stoves, a public consultation has been planned two months after the distribution to collect the beneficiaries' opinions and recommendations. This consultation was scheduled for Wednesday 07 December 2022. Participants have been invited mainly by telephone and the venue for the consultation will be in Kpélé Adéta.

At the end of the consultation, 100 improved cookstoves are distributed free of charge to the beneficiaries. The list of beneficiaries is attached to this report.

Contact details are provided on the contract signed by the end-users who acquire a cookstove, to ensure that they can contact project proponent at any time.



Figure 2: Symbolic handover of the stove to a beneficiary



Figure 3: Attendees at a LSC meeting

2.3 Environmental Impact

No negative environmental impacts have been identified for the project and an environmental impact assessment (EIA) is not required for the project.

2.4 Public Comments

The project was open for public comment from 16/01/2023 to 15/02/2023 and did not receive any comments.

2.5 AFOLU-Specific Safeguards

Not applicable as this project in a non-AFOLU project

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1

Tools: "TOOL30: Calculation of the fraction of non-renewable biomass" Version 4.0

3.2 Applicability of Methodology

Applicability criterion VMR0006 v1.1	How the project complies
Project activities shall be implemented in domestic premises or in community-based kitchen	The proposed project involves deployment of ICS only in households in Togo, and therefore meets this applicability condition
The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot;	TCHA-TCHA Business and GLOIRE A DIEU stoves installed under this project are single pot wood cookstoves that have an efficiency of 31.82% as per the Water Boiling test. The efficiency of the stoves is above 25%. Therefore, this applicability condition is met.
Both 'Projects' and 'Large Projects' can use this methodology	This project is a large project. The methodology is applicable to both project and large project. Therefore, this applicability condition is met.

Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics;	The source of the biomass is from non-renewable sources in Togo; The biomass traditionally utilized is a mixture of non-renewable biomass from natural woodlands, charcoal produced locally and sold via retail transporters in small quantities (one bag of charcoal or less than 100 kg of wood) to urban centers, especially by bicycle or on the head. And they sell directly to retail vendors or consumers. Based on ReddTogo report, Biomass energy alone represents 71% of the national energy balance and more than three quarters of domestic energy sources still come today from the use of plant biomass.⁷ The country has anticipated by initiating since 1980 a policy of replacing part of the consumption of wood energy by gas butane. However, the use of gas has seen very low penetration due to gas prices and equipment lockout which, for the majority of urban households, are too high. That is why the harvesting of non-renewable biomass is commonplace in Togo and can be reasonably concluded that has been the case since 31 December 1989. Between 1990 and 2020, Therefore, it can be proved that non-renewable biomass has been used in Togo since 1989
For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel	Not applicable. The ICS are introduced as energy efficiency measure to replace baseline stoves and reduce the use of non-renewable biomass for combustion

⁷ file:///C:/Users/33666/Downloads/Rapport_Etude_Bois-Energie_REDD_Togo_2017.pdf

should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded	
The CDM-PDD or CDM-PoA-DD/CPA-DD shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo).	A record keeping system and unique identification of the project device will be used. This will include username, user identification number and address/location of the user's house and distribution date. The record keeping system will ensure that each ICS can be traced to project to avoid double counting
The CDM-PDD or CDM-PoA-DD/CPA-DD shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	The project proponent will sign an end user agreement with individual household/user to ensure only PP can claim the emission reduction. A default distribution agreement for end users including the provision that emission reductions generated by the stove are owned by the PP will be provided for each user.

3.3 Project Boundary

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Yes	Significant source of emissions
	CH ₄	Yes	Significant source of emissions
	N ₂ O	Yes	Significant source of emissions
	Other	No	No other source identified
Project	CO ₂	Yes	Significant source of emissions
	CH ₄	Yes	Significant source of emissions

Source		Gas	Included?	Justification/Explanation
	Combustion of wood fuel for cooking	N ₂ O	Yes	Significant source of emissions
		Other	No	No other source identified

The project activity will be located solely within from the boundaries of Togo.

3.4 Baseline Scenario

The applied methodology VMR0006 Methodology for Installation of High Efficiency Firewood Cookstoves v1.1, quantifies emission reductions as a function of the reduction in the amount of non- renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves.

For Togo, the share of biomass-energy reaches 71% of the national final consumption with 90% of the population consuming wood energy, based on the ReddTogo study conducted in 2017⁸. Moreover, we know that only 10% of the population has access to clean cooking energy⁹. In Togo, as in other countries in the subregion, biomass (firewood, charcoal, other products and vegetable waste) is the most important household energy source, accounting for 71%¹⁰ of the country's total energy consumption. 93% of households use wood energy to cook food (58.% for firewood and 35% for charcoal). Therefore, the baseline scenario is the usage of firewood or charcoal in traditional low efficiency charcoal/firewood stove or three-stone for cooking in kitchens of households.

The baseline scenario is identified as the use of woody biomass from non-renewable sources in traditional charcoal and three stone cooking fires.

Therefore, the baseline scenario is the continued use of charcoal or firewood in inefficient cookstoves by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

3.5 Additionality

The methodology uses activity method for the demonstration of additionality.

⁸file:///C:/Users/33666/Downloads/Rapport_Etude_Bois-Energie_REDD_Togo_2017.pdf

⁹ [Togo | Tracking SDG 7 \(esmap.org\)](http://www.esmap.org/togo-tracking-sdg-7)

¹⁰

http://www.globalbioenergy.org/fileadmin/user_upload/gbep/docs/2018_events/TOGO_1st_meeting/MERF_Bois_%C3%A9nergie_Togo.pdf

Activity Method

Step 1: Regulatory Surplus

There is no mandated government programme or policy in the present host country (Togo, same will be verified for each new country included) of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law, statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework. Households may only participate voluntarily in this project. The proposed project is a voluntary coordinated action by Sequoia

Step 2: Positive List

This project activity meets all the applicability conditions of the methodology, as demonstrated above, but also the conditions above:

1. Where the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional => Project proponent will distribute the cookstoves at no cost to the end user and has no other source of revenue other than the sale of GHG credits,
2. Project activities that are implemented as part of government schemes or are supported by multilateral funds cannot be considered additional even if the stoves are distributed free of cost or at a highly subsidized rate and hence are not eligible to use this methodology. => Project activity is not implemented as part of government schemes nor supported by multilateral funds.

Hence the project activity is deemed additional.

Step 3. Project Method

Not applicable as the project is distributed at zero cost.

3.6 Methodology Deviations

There is no methodology deviation.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The approved methodology does not account for baseline emissions separately but quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass

fuel consumption in the efficient project stoves as compared to baseline stoves (see section 4.4. below)

4.2 Project Emissions

The approved methodology does not account for project emissions separately but quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves (see section 4.4. below).

4.3 Leakage

Leakage shall be considered as default 0.95 in accordance with Section 5.4 of the approved methodology.

4.4 Net GHG Emission Reductions and Removals

The improved cookstove is introduced as energy efficiency measure in the project, therefore equations 1 and 2 of the methodology will be applied to calculate the net GHG emission reductions.

$$ER_y = \sum_i \sum_j ER_{y,i,j}$$

i	=	Indices for the situation where more than one type/model of improved cookstove is introduced to replace three-stone fire
j	=	Indices for the situation where there is more than one batch of improved cookstove of type i
ER_y	=	Emission reductions during year y in t CO ₂ e
$ER_{y,i,j}$	=	Emission reductions by improved cookstove of type i and batch j during year y in t CO ₂ e

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{wood\ fuel} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95$$

Where:

$B_{y,savings,i,j}$	=	Quantity of woody biomass that is saved in tonnes per improved cook stove of type i and batch j during year y
$f_{NRB,y}$	=	Fraction of woody biomass that can be established as non-renewable biomass (fNRB)
$NCV_{wood\ fuel}$	=	Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)

$EF_{wf,CO2}$	=	CO2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel,
$EF_{wf,non\ CO2}$	=	Non-CO2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO2/TJ
$N_{y,i,j}$	=	Number of improved cook stoves of type i and batch j operating during year y

$$B_{y,savings,i,j} = B_{old} \times (1 - (\eta_{old} / \eta_{new,i,j}))$$

$$\text{With } \eta_{new,i,j} = \eta_P \times (DFn)^{y-1} \times 0.94$$

Where:

B_{old}	=	Quantity of woody biomass that is saved in tonnes per improved cook stove of type i and batch j during year y
η_{old}	=	Efficiency of baseline cookstove
$\eta_{new,i,j}$	=	Efficiency of the improved cookstove type i and batch j determined through water boiling test (WBT) during year y Alternatively, efficiency may be determined using Equation 4
η_P	=	Efficiency of project stove (fraction) at the start of project activity,
$(DFn)^{y-1}$	=	Discount factor to account for efficiency loss of project cookstove per year of operation(fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards Alternatively default value of 0.99 efficiency loss per year can be considered
0.94	=	Adjustment factor to account for uncertainty related to project cookstove efficiency test

Parameter	Value	Unit	Description of the parameter	Source/reference/justification
$EF_{wf,CO2}$	112	tCO2/TJ	CO2 emission factor for the use of wood fuel in baseline scenario	IPCC default for wood fuel,
$EF_{wf,non\ CO2}$	26.23	tCO2/TJ	Non-CO2 emission factor for the use of wood fuel in baseline scenario	IPCC default for wood fuel

$N_{y,i,j}$	3,444	Cookstoves	Number of cookstoves distributed and operational the first year.	Ex-ante estimation of operating cookstoves
$NCV_{\text{wood fuel}}$	0.0156	TJ/tonne	Net calorific value of the non-renewable woody biomass that is substituted or reduced	IPCC default factor for wood
η_{old}	14% (Ex-ante)	fraction	Efficiency of baseline stoves	Ex-ante estimated Weighted average efficiency.
$\eta_{\text{new},i,j}$	31.82%(1 st year)	fraction	Efficiency of the improved cookstove type i and batch j. Efficiency may be determined using Equation 4	Calculated
f_{NRB}	74.20%	Fraction	Fraction of woody biomass that can be established as non-renewable biomass	Calculated
B_{old}	3.05	tonnes	Annual quantity of woody biomass that would have been used in the absence of the project activity (in tonnes per device) to generate useful thermal energy equivalent to that provided by the improved cook stove	Calculated (section 5.1)
μ_y	10%	fraction	Factor to account for the continued use of baseline stoves	Ex-ante estimation to be monitored ex-post
$B_{\text{old adjusted}}$	2.745	tonnes	Annual quantity of woody biomass that would have been used in the absence of the project activity (in tonnes per device) to generate useful thermal energy equivalent to that	calculated

			provided by the improved cook stove taking into consideration the continued use of baseline stoves	
η_p	31.82%(1 st year)	Fraction	Efficiency of project stove (fraction) at the start of project activity	Obtained from WTB (SOSET report)
$(DF_n)^{y-1}$	0.99	-	Discount factor to account for efficiency loss of project cookstove per year of operation(fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered	Default value from methodology
$B_{y,savings,i,j}$	1.54(1 st year)	tonnes	Quantity of woody biomass that is saved in tonnes per improved cook stove of type	Calculated
μ_y	90%	Fraction	Adjustment to account for any continued use of pre-project devices during the year y	Ex-ante estimation
L	95%	fraction	Leakage adjustment factor	Methodology default
ER	7,266	tCO ₂ e	Emission reduction for first year	Calculated as per methodology equation

The quantity of firewood consumed in absence of project activity (**Bold**) shall be determined using an estimation of average annual consumption of firewood per household which may be derived using any of the following options:

- (a) Historical Data. Project proponent shall ensure that the relevance of data is appropriately justified for the target population and is the latest available data from credible source(s).
- (b) Baseline Survey of Local Usage. Project proponent shall carry out a survey of usage prior to implementation of the project activity following the sampling approach described in the latest version of CDM document Sampling and surveys for CDM project activities and programme of activities. Alternatively, the project participant may follow the simple random sampling approach and the minimum sample size should be determined as per the following guidelines:
- Project target population < 300: Minimum sample size 30
 - Project target population 300 – 1000: Minimum sample size 10% of group size
 - Project target population > 1000: Minimum sample size 100

This simplified approach may also be used for determining minimum sample size for parameters listed under Sections 9.1 and 9.2 in which case it is not requisite for the sample size to meet confidence/precision requirements

- (c) Minimum Service Level. Where historical data or a baseline survey has not been conducted, a default value of 0.5 ton/capita/year may be considered as the baseline biomass consumption. Household size shall be determined using credible references/literature or target population specific surveys. The survey shall be conducted as per guidelines outlined in option (b) above

The equations below shall be used for calculating biomass consumed in absence of project activity in case more than one project stove is used in household:

$$\text{Bold}_{i,j} = \text{Bold}_{HH} + N_{d,HH}$$

$$\text{Bold}_{HH} = \text{Bold}_p \times N_{p,HH}$$

Bold _p	=	Annual quantity of woody biomass that would have been used per person in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices (tonnes/person/year).
N _{p,HH}	=	Average number of person per households
N _{d,HH}	=	Number of project devices per household
Bold _{HH}	=	Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices (tonnes/household/year)

As per section 8.4 of the applied methodology, Bold can be estimated using baseline survey data (Option b). This option is chosen for the determination of Bold. Therefore, above parameters are determined using baseline survey results data.

The consumption of woody biomass is retrieved from most recent, credible and project specific data available from credible source.

Determination of Bold,p.

The baseline study have been conducted by LTA conseils in 2022. The study (page 10) conducted¹¹ underlined a charcoal consumption of 42.36 kg per HH/month hence a consumption of 3,050 kg/year/hh and a firewood consumption of 390 kg/HH per month, hence a consumption of 4,687 kg/year/hh for an household size of 5. The lowest value (3,050 kg/HH/year or 610kg/capita/year) is used conservatively. This value is also found conservative with regards to the expired CDM standardized baseline for cookstoves in Togo¹² underlining an average (rural and urban) consumption of 925 kg of biomass per person per year

Therefore, data from this study as most recent will be used for Bold,p.

Determination of Np,HH

Based on the baseline report conducted by LTA for Togo the average household size is 5.

Thus this value was selected for households size estimation.

Determination of Nd,HH

As the project is expected to distribute only one ICS per household, therefore the Nd,HH value is 1

Determination of Bold, HH

Bold, HH = 3.05 tonnes per household and per year

Determination of Bold,i,j

Bold,i,j = Bold,HH ÷ Nd,HH = 3.05 / 1 = 3.05 tonnes

All the parameters are resumed in the table below:

Parameters	Value	Source
Bold,p	0.610tonnes/person/year	Calculated
Nd,HH	1	Ex-ante estimation
Np,HH	5	department of Economic and Social Affairs of the United Nations
μ _y	90%	Ex-ante estimation

¹¹ [LTA Conseils Baseline survey](#)

¹² https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20170223112453889/ASB0033_TSB0010.pdf

<i>Bold,HH</i>	3.050	Calculated as per methodology
<i>Bold,HH adjusted</i>	2.745	Calculated as per methodology

Determination of efficiency of ICS during year y

As per equation 5 of methodology applied $\eta_{new,i,j} = \eta_P \times (DF_n)^{y-1} \times 0.94$

The ex-ante estimated efficiencies are therefore:

Year	$\eta_{new,i,j}$
2022	31.82%
2023	29.9%
2024	29.6%
2025	29.3%
2026	29%
2027	28.7%
2028	28.4%
2029	28.2%

Determination of Bysavings. $B_{y,savings,i,j} = \text{Bold} \times (1 - (\eta_{old} / \eta_{new,i,j}))$

Year	Bold adjusted(tonnes)	$\eta_{new,i,j}$	η_{old}	$B_{y,savings,i,j}$ (tonnes)
2022	2.745	31.82%	14%	1.54
2023	2.745602	29.9%	14%	1.47
2024	2.745602	29.6%	14%	1.45
2025	2.745602	29.3%	14%	1.44
2026	2.745602	29%	14%	1.43
2027	2.745602	28.7%	14%	1.41
2028	2.745602	28.4%	14%	1.40

2029	2.745602	28.2%	14%	1.39
------	----------	-------	-----	------

Determination of fNRB

fNRB was estimated following tool 30 methodology : The values obtained is 74.20%. Main results for calculation are provided below¹³.

Parameter	Value	Source of data
NRB	3,259,481.68 t/yr	Calculated
RB	1,132,602.57 t/yr	Calculated
H	4,392,084.26 t/yr	Calculated
MAIforest,i	1.52 t/ha/yr	Calculated
MAIother	1.52 t/ha/yr	Calculated
Fforest	1,209,270 ha	FRA2020 country report
FnrB	74.20 %	Calculated

Comparison of calculated fNRB with the values for fNRB reported in relevant scientific literature:

The achieved fNRB was compared with those available on literature mentioned by tool 30 version 4 (Bailis, R.; Drigo, R.; Ghilardi, A. & Masera, O. (2015). The carbon footprint of traditional woodfuels. Nature Climate Change, 5(3), pp. 266–272) however, the fnrB from the literature was not used for the following reasons:

The study was made in 2015, which means 8-9 years ago. And just as for other parameters, a value that old is not taken into account, unless it cannot be found via another source or calculated. data used to calculate fNRB results from country-driven official reports and statistics, just as the global forest resources assessment, which data is provided by a well-established network of officially nominated national correspondents.

Data used for wood consumption and local supply/demand is from 2009, which makes these value non applicable to the project as the values at this time may not represent actual situation. Furthermore, since 2015, the Togolese population continued to increase (more than a million more inhabitants) meaning a constant increase of wood fuel consumption. As per latest available data from Forest resource assessment¹⁴ forest area continues to decrease (-59,190 ha between 2000 and 2020) with a

¹³ Full and detailed calculation is available in the ER calculation sheet

¹⁴ <https://fra-data.fao.org/assessments/fra/2020/TGO/sections/extentOfForest>

deforestation rate of 4,960 ha/year. In the same time, biofuel is still the main fuel source used by households for energy¹⁵.

The study conducted by Bailis and al, is based on the WISDOM approach that rely on extensive data research, geo-spatial analysis, hypothesis based on researchers' knowledge, while the tool 30 is a more accessible way of measuring the fNRB, making it difficult to compare both results, although the result obtained by project developers seems more aligned with the latest value of fNRB approved by CDM Board (valid from 2013 to 2018) and estimated at 97% for Togo.

Also, the study explains that "For analytical purposes, the continuous map resulting from the weighted interpolation analysis is segmented into buffers; cities with high demand produce wide woodshed buffers and cities with low demand produce narrow buffers, which simulates the territory under pressure from urban and high-deficit rural areas. Woodsheds within each national boundary are defined by using zonal statistics to calculate the supply/demand balance of each buffer" However, buffering points can underestimate or overestimate woodshed availability in each zone, based on statistics used. As explained in the study, this approach assumes optimal harvesting of DEB (dendro-energy biomass). It does not reveal actual harvesting patterns" Thus it could lead to inaccurate values for each zone and buffer.

On the other hand, data used to calculate fNRB results from country-driven official reports and statistics, just as the global forest resources assessment, which data is provided by a well-established network of officially nominated national correspondents.

The FNRB is also compared to:

Project name	Link to the project	fNRB value	Comments
Installation of high efficiency wood burning cookstoves in Togo	https://registry.terra.org/app/projectDetail/VCS/4160	87%	Value higher than calculated fNRB
EMPOWERING COMMUNITIES THROUGH IMPROVED COOKSTOVES	https://registry.terra.org/app/projectDetail/VCS/3881	87%	Value higher than calculated fNRB

Conclusion:

The fNRB assessment methodology used by Bailis and al; does not seems to be applicable to this project, hence the results cannot be used for the project. The report does not use the CDM Tool 30 to assess the fNRB although this is the tool proposed by both AMS II-G and VMRO006 (fNRB default values from CDM)

¹⁵ <https://au-afrec.org/togo>

to assess the value of fNRB. Moreover due to the fact that these values from the study rely on really old data, with no guidance from VERRA on how to reassess fNRB based on WISDOM methodology with updated data, project developer tend to believe that although many fNRB assessments methods exist (tool 30, WISDOM, MoFuSS) with both their pro and cons, the most used and accepted so far way of assessing the fNRB (cdm tool 30) can be used and value assessed by project developer can be applied.

The fNRB, is anyway found to be more conservative than the other ICS projects in Togo hosted in VERRA registry.

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
18/09/2022-31/12/2022	2,089	0	0	2,089
2023	360,865	0	0	360,865
2024	715,365	0	0	715,365
2025	797,553	0	0	797,553
2026	878,052	0	0	878,052
2027	1,043,823	0	0	1,043,823
2028	1,171,735	0	0	1,171,735
01/01/2029-30/09/2029	826,638	0	0	826,638
Total	5,796,120	0	0	5,796,120

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	Bold		
Data unit	tonnes/year		
Description	Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices.		
Source of data	Historical data		
	Number of person by households	5	Togo baseline report available on page 10 at <u>LTA Conseils Baseline survey</u>
	Biomass consumption by capita (kg/day)	0.610	Togo baseline report available on page 10 at <u>LTA Conseils Baseline survey</u>report underlined : 1) a consumption of household size of 5, 2) an average charcoal consumption of 42,6kg of charcoal/hh/month hence an equivalent of 3,067 tonnes (42.6*12*6) of biomass/HH/year or 613 kg/capita/year. 3) An average consumption of 390 kg of wood/HH/month hence a consumption of 4,687 tonnes/HH/year or 937 kg/capita/year To be conservative the lowest value of biomass consumption (613kg) is used instead of the mean (charcoal and firewood) or the weighted average. Please refers to calculation sheet tab"Determination od Bold and Nold) for detailed calculation)
Value applied	3.050		
Justification of choice of data or description of measurement methods and procedures applied	Based on the lowest value of biomass consumed by person/year times the average numbers of people per household		
Purpose of Data	Calculation of Quantity of woody biomass that is saved in tonnes per device		
Comments			

Data / Parameter	$\eta_{old,i,j}$
Data unit	Fraction

Description	Efficiency of the system being replaced (Baseline Stoves)										
Source of data	ex-ante estimate based on the weighted average of charcoal and firewood consumption.										
Value applied	14%										
Justification of choice of data or description of measurement methods and procedures applied	Option (e) of the methodology page 19 is chosen : (e)If more than one type of baseline device is being replaced in the project region, weighted average values (taking the amount of woody biomass consumed by each device as the weighting factor) shall be used) <table border="1"> <tr> <td>Charcoal consumption in biomass equivalent (tonnes/HH/year)</td><td>3,050</td></tr> <tr> <td>Firewood consumption in biomass equivalent (tonnes/HH/year)</td><td>4,687</td></tr> <tr> <td>Coal pot efficiency (%)</td><td>20%</td></tr> <tr> <td>Firewood efficiency (%)</td><td>10%</td></tr> <tr> <td>Weighted average efficiency</td><td>14%</td></tr> </table> Refers to the tab "Determination of Bold and Nold" in the calculation sheet for detailed calculation	Charcoal consumption in biomass equivalent (tonnes/HH/year)	3,050	Firewood consumption in biomass equivalent (tonnes/HH/year)	4,687	Coal pot efficiency (%)	20%	Firewood efficiency (%)	10%	Weighted average efficiency	14%
Charcoal consumption in biomass equivalent (tonnes/HH/year)	3,050										
Firewood consumption in biomass equivalent (tonnes/HH/year)	4,687										
Coal pot efficiency (%)	20%										
Firewood efficiency (%)	10%										
Weighted average efficiency	14%										
Purpose of Data	Calculation of Quantity of woody biomass that is saved in tonnes per device (calculation of emissions reduction)										
Comments	--										

Data / Parameter	n_p
Data unit	Fraction
Description	Efficiency of the project stove at the start of project activity
Source of data	Manufacturer's specification based on Water boiling Test conducted.
Value applied	0.3182
Justification of choice of data or description of measurement methods and procedures applied	Parameter determined ex-ante

Purpose of Data	Calculation of baseline emissions
Comments	This efficiency is for the models distributed so far. Np may be updated to reflect efficiency of all the new types of stoves that will be included.

Data / Parameter	fNRB,y
Data unit	Percentage
Description	Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass
Source of data	Fraction of non-renewable biomass is calculated as per the fNRB calculation TOOL30 v4: Calculation of the fraction of non-renewable biomass
Value applied	74.20%
Justification of choice of data or description of measurement methods and procedures applied	Any changes/ revisions to this value would be monitored and if new default values are approved by the CDM EB the value of the fraction of NRB used will be updated.
Purpose of Data	Calculation of baseline emissions
Comments	This parameter shall remain fixed for the monitoring periods.

Data / Parameter	NCV _{biomass}
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted
Source of data	Default as per applied approved methodology VMR0006
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	Default as per applied approved methodology VMR0006
Purpose of Data	Calculation of baseline emissions
Comments	This parameter shall remain fixed for the monitoring periods.

Data / Parameter	EF _{wf, non CO2}
Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	Default as per applied approved methodology VMR0006
Value applied	26.23
Justification of choice of data or description of measurement methods and procedures applied	Default as per applied approved methodology VMR0006
Purpose of Data	Emission reduction calculations
Comments	This parameter shall remain fixed for the monitoring periods.

Data / Parameter	EF _{wf,CO2}
Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	Default as per applied approved methodology VMR0006
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	Default methodology value.
Purpose of Data	Emission reduction calculations
Comments	This parameter shall remain fixed for the monitoring periods.

5.2 Data and Parameters Monitored

Data / Parameter	N _{y,i,j}
Data unit	Number

Description	Number of project devices of type i and age a that are operating in year y	
Source of data	ICS users' survey	
Description of measurement methods and procedures to be applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and programme of activities.	
Frequency of monitoring/recording	At least once every two years	
Value applied	Year	Nyi
	18/09/2022-31/12/2022	3,444
	2023	180,000
	2024	360,000
	2025	405,000
	2026	450,000
	2027	540,000
	2028	612,000
	01/01/2029-17/09/2029	612,000
Monitoring equipment	Monitoring surveys	
QA/QC procedures to be applied	The project implementor shall maintain a distribution record to calculate this parameter	
Purpose of data	Calculation of baseline emissions	
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value. For ex-ante it is estimated that 90% of the stoves will be operational	
Comments	--	

Data / Parameter	$\eta_{\text{new},y,i,j}$	
Data unit	Fraction	
Description	Efficiency of the improved cookstove type i and batch j	
Source of data	Water Boiling Test	
Description of measurement methods and procedures to be applied	To adopt Option V given in the methodology: “Efficiency of the improved cookstoves to be estimated using equation 5 above where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored”	
Frequency of monitoring/recording	Annual	
Value applied	Year	Efficiency
	18/09/2022-31/12/2022	31.82%
	2	29.9%
	3	29.6%
	4	29.3%
	5	29%
	6	28.7%
	7	28.4%
	01/01/2029-17/09/2029	28.2%
Monitoring equipment	n/a	
QA/QC procedures to be applied	n/a	
Purpose of data	Calculation of baseline emissions	
Calculation method	As per equation 5 of VMR0006 i.e. $\eta_{\text{new},i,j} = \eta_P \times (DFn)^{y-1} \times 0.94$	

Comments	--
Data / Parameter	μy
Data unit	Fraction
Description	Adjustment to account for any continued use of pre-project devices during the year y
Source of data	Monitoring survey
Description of measurement methods and procedures to be applied	This parameter should be monitored using one of the following methods: a) If the baseline cookstoves are decommissioned and no longer used, as determined by the monitoring survey its value is 0 and Bold, adjusted is equal to Bold. b) If both the improved cookstove and baseline cookstoves are used together then surveys shall be conducted to record the average continued operation of baseline cookstoves in a sample of households. The surveys should be designed to capture the cooking habits and stove usage of households in the region, including quantification of use of baseline cookstoves, by formulating questions and/or collecting evidences to determine the frequency of usage of both the improved cookstoves and baseline cookstoves. For example, if there were 3 baseline cookstoves in a household and it was determined during the survey that use of one of them continues during the crediting period then a conservative adjustment factor of 0.33 is applied to Bold. Another example would be the case where there was only one baseline cookstove per household and its use during the project period continues along with the improved cookstove to meet 25% of the cooking needs of the household in which case the adjustment factor will be 0.25. Another example would be to interview the household and have them estimate the time of usage of the baseline cookstoves and improved cookstove on an average day.
Frequency of monitoring/recording	Once every 2 years
Value monitored	10% (ex-ante estimation)
Monitoring equipment	Monitoring surveys
QA/QC procedures to be applied	A 95 /10 confidence / margin of error shall be achieved for the sampling parameter irrespective of annual /biennial monitoring frequency as per para 22 of Standard: Sampling and surveys for

	CDM project activities and programmes of activities, Version 09.0. In the case the desired precision is not met, lower bound values shall be used against repeating the survey to determine the operational fraction of stoves.
Purpose of the data	Calculation of baseline emissions
Calculation method	n/a
Comments	The proportion of continued use of pre-project stoves is a monitored parameter, i.e. only available ex-post. For ex-ante purposes, $\mu_y = 90\%$ can be considered. For the subsequent monitoring sessions of the same project, the previously monitored values of μ shall be used for ex-ante purposes.

Data / Parameter	Life Span
Data unit	Number of years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency
Source of data	Manufacturer's specification
Description of measurement methods and procedures to be applied	Lifetime determined as per laboratory test
Frequency of monitoring/recording	Once at the time of project stove installation
Value monitored	5
Monitoring equipment	n/a
QA/QC procedures to be applied	n/a
Purpose of the data	Calculation of baseline emissions
Calculation method	n/a

Comments	After their lifetime of 5 years (starting from their date of installation) , the stoves will be automatically removed from the database, and no emissions reductions will be requested for theses cookstoves even if they are still operational.	
Data / Parameter	Date of commissioning of batch j	
Data unit	Date	
Description	To establish the date of commissioning, the Project Participant may opt to group the devices in “batches” and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch.	
Source of data	Project database	
Description of measurement methods and procedures to be applied	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch	
Frequency of monitoring/recording	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch	
Value monitored	Batch	Date of commissioning
	2022	18/09/2022-31/12/2022
Monitoring equipment	n/a	
QA/QC procedures to be applied	n/a	
Purpose of the data	Calculation of baseline emissions	
Calculation method	n/a	
Comments	--	

5.3 Monitoring Plan

The monitoring plan is designed to monitor the parameters listed in Section above, which are required for calculation of the actual GHG emission reduction achieved by the project activity using ex post sampling survey. The share of operating stoves and the continued use of pre-project

devices will be determined based on sampling procedures as outlined below. The PP will be responsible for conducting the sampling surveys and maintaining a database with all operating stoves.

No monitoring for leakage is required, as the Emission reduction will be multiplied by a factor of 0.95 to account for leakage.

Based on CDM Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 09.0, the sampling plan is the following: (a) Sampling design: Due to the large number of ICS envisioned to be distributed under the project, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling shall be undertaken for each batch of ICS. Samples will be drawn from ICS recorded in the project database administered by the project proponent.

Objectives and Reliability Requirements

The objective is to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of the crediting period. As per Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities: “When biennial inspection is chosen a 95 per cent confidence interval and a 10 per cent margin of error shall be achieved for the sampling parameter. On the other hand, when the project proponent chooses to inspect annually, a 90 per cent confidence interval and a 10 per cent margin of error shall be achieved for the sampled parameter”. Therefore, for parameters monitored annually, sample size should permit to achieve a 90/10 precision, and for parameters monitored biennially, the sample size should be determined to achieve a 95/10 precision.

Monitored Parameters: The project involves introduction of intermediate improved cook stoves as energy efficiency measure:

No.	Monitoring parameters	Sampling parameters	Parameter type	Monitoring frequency
1	$N_{y,i,j}$ Number of project devices of type i and batch j operating in year y	Proportion of ICS still in operation	Proportion	Biennially

2	$\eta_{new,y,i,j}$ Efficiency of the device of each type i and batch j implemented as part of the project activity	Thermal Efficiency of operational ICS	Mean value	Annually (The project opts to determine the efficiency using the equation given by methodology, therefore it is not required to monitor this parameter via sampling survey.)
3	μ_y Adjustment to account for any continued use of pre-project devices during the year y	Adjustment factor	Proportion	Biennially

Target Population

The target population will be the complete number of devices (ICS) distributed under the project activity.

Sampling Method

The project involves distribution of ICS throughout the project area thereby replacing traditional cooking devices in the selected households. The population is heterogeneous in nature i.e., common technology with similar operating characteristics but spread across the eastern parts of the country.

The populations of each batch will be combined, the sample size is determined, and a single survey will be undertaken to collect data. To ensure the survey result is representative of the entire population, the dissimilarity (such as ICS type, vintage and project area in which they are operating) within the project will be taken into account in the sample size calculation. Samples will be drawn by using the random number generator.

Monitoring organization:

The project operator Sequoia Togo takes the responsibility of the monitoring plan implementation, supervision of the monitoring process, periodic users surveys, data collection and recording, and reporting.

The information collected will be stored in the electronic database available with PP and it will be updated on ongoing basis by trained staff. The database will be sortable by the information

collected as per Registration Process and will be made available to the VVB at verification. The PP will verify accuracy and completeness and confirm that there is no double entry of serial numbers in the database. The PP will identify any discrepancy and the correct information will be entered into the database.

All partners will be required to conform with project design and monitoring systems designed which will cover the above-mentioned role and responsibilities.

All technical staff responsible for installation and maintenance of the stoves will be trained in terms of the understanding the requirements of VCS on the monitoring system. The technical, operational and maintenance trainings provided for the personnel will be described in each monitoring report.

Below, the team in charge of the monitoring:

Role	Responsibilities	Identity
Monitoring manager	- In general, ensure that the project follow the monitoring plan - Ensure that the equipment and measurements are in line with the measurement methods, recording frequency and archiving approaches in monitoring plan - Ensure that monitoring data collected is consolidated and entered in electronic database - Ensure that monitoring team receives proper training - Ensure a coherent and standard monitoring report for the verification	SIRANGI Rajesh
Monitoring surveys	- Conduct the monitoring surveys, and follows up visits	MOLAGUI Kokou Hervé SOA Milodgue N'ni AMEDONOU Agbekossi Kossi
Data manager	- Collect monitoring data. - Enter data in electronic database and archive hardcopies.	KIDJA Jocelyn Ekoe

Consultant	- Carry out sample size determination and emission reduction calculations	AERA Group
------------	---	------------